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Date: 3/12/25

Element K: Reflection on the design project

The enrichment project undertaken to satisfy the needs of the Nashville Zoo's elands required a deliberate and thoughtful process, balancing technical requirements against practical constraints. The task was to create a durable, safe, and budget-friendly toy hanger system that would stimulate Loki, Olive, and Violet, three elands with distinct physical needs, without compromising zookeeper safety standards. The project's success was based on the resolution of the problem, organized search for solutions, and iterative prototyping based on interdisciplinary advice. Every phase of the project required serious consideration of the welfare of the animals, structural integrity, and functional operability, providing useful information on the nature of designing in captive wildlife enclosures.

The demand for the problem stemmed from the boredom of the elands in their holding environment, which manifested itself as pacing, stereotypy, and aggression, indications of inadequate mental stimulation. Elands are large, intelligent herbivores and require environments that allow for natural challenge, including exploratory play and foraging. Existing infrastructure was not capable of suspending enrichment toys at heights suitable for horn interaction, which is a critical component of their natural behavior. The structure had to hold toys up to 100 pounds at 10-14 feet to be within reach of the elands without risk of entanglement. Safety was the priority, especially considering the zookeepers' preference not to enter the enclosure. This required a remotely controlled system that was strong enough to hold the force of a 1,300-pound male eland but simple enough for one keeper to operate. The \$200 budget requirement created an added layer of complexity in material choice, requiring creativity in finding a balance between cost and longevity.

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Conceptualization began with an examination of industrial cranes, which informed the design of the miniaturized manual crane. Brainstorming sessions considered motorized mechanisms, pulley systems, and materials, but safety and budget concerns soon focused on manual operation. A scoring matrix evaluating concepts against requirements of safety, durability, and user interest indicated that hand systems with levers or hand cranks were superior to automated ones. Safety was the top priority, excluding motorized components with entanglement risks or breakdown. Durability tests favored steel beams and concrete bases, but budget considerations necessitated a recourse to pressure-treated wood and nylon cords. This phase highlighted the importance of flexibility, early suppositions about mechanisms and materials having to yield to budget constraints without compromising on vital functionality.

Prototyping involved vast experimentation with structural configurations and material choices. Initial prototypes utilized 3D-printed components to prototype load-carrying joints and pulley systems, and engineer and contractor feedback assisted in refining the design. A civil engineer's recommendation that the vertical beam be cast in concrete significantly enhanced stability, eliminating concerns regarding land contact-generated lateral forces. Rebar in the concrete base further stabilized the structure, while steel cables and turnbuckles replaced diagonal wooden supports to reduce weight and increase tensile strength. These developments emphasized the value of external expertise in identifying vulnerabilities, e.g., stress points at joint connections or wood splitting due to torsion. The repetitive process revealed that simplicity is often an enhancer of reliability; for instance, replacing complicated bracket systems with steel cables minimized assembly and maintenance.

Material science also played a pivotal role in rendering the prototype feasible. Decisions to authenticate nylon cord load-bearing strength and pulley system friction loads

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were made to prevent loss of toy or rope degradation. Substitution with wooden 4x4s for Unistrut metal framing was an excellent example of cost-savings flexibility, but required the need to closely review wood grain direction and bolt location to prevent structural failure. The height of the end design was elevated to 16 feet based on zookeeper feedback that emphasized the need for toys to have adequate clearance above the eight-foot-tall fence with a notable buffer, ensuring ease of maintenance. Such enhancement demonstrated how consistent dialogue with end-users, zoostaff, regulated pragmatic realities, such that the design had the advantage of a real on-the-ground cycle of operations.

Learning from the project shows a range of essential takeaways. To begin with, setting clear-cut, ranked requirements early during design is priceless. Safety and budget constraints acted as a brake, steering the decision away from promising but non-viable options. Second, interdisciplinary collaboration added depth to the solution; contributions from engineers, builders, and zookeepers took abstract models and turned them into functional designs. Third, prototyping is not only a test phase but an investigative stage; each iteration uncovers more nuanced mistakes, from material limitations to ergonomic oversights. Finally, the constraint on resources generated ingenuity, illustrating that constraints stimulate



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creativity more than they impede it. On future projects, the lessons affirm the need to involve stakeholders, ongoing prototyping, and a willingness to adapt in reaction to unforeseen issues. The Eland toy hanger project, humble as it was in scale, though perhaps it was, encapsulates the process of engineering design: human and in this case, animal, problem-solving through the application of empathy, discipline, and flexible thought.

